

Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 4: Expressions Using Letter-Numbers

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	4 — Expressions Using Letter-Numbers
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps at this level: charging an 'included first item' twice, distributing a coefficient (or a leading minus) over only the first term of a bracket, merging unlike terms or terms of different powers, dropping or sign-flipping a constant, squaring a sum as if it were the sum of squares, applying a negative sign before versus after squaring, mistaking $2n$ for $n+2$ or $3a$ for a -cubed, and reading a pattern's first value as its common difference. Keep every sign exactly as given; only LIKE terms combine; this chapter builds and simplifies expressions and reads off rules — it never solves equations. Do not write on this paper — mark answers on your answer sheet.

1. A courier charges a fixed 70 rupees that already covers the first 3 kilometres, then 18 rupees for each kilometre after the third. A second courier has no fixed charge but bills 25 rupees per kilometre from the very first. For a trip of d kilometres (d greater than 3), the AMOUNT by which the second courier costs MORE than the first, in rupees, simplified, is:

- (A) $7d + 16$ (B) $43d - 16$
(C) $7d - 16$ (D) $7d - 70$

2. Evaluate $2a^2 - 3ab + b^2$ when $a = -2$ and $b = 5$.

- (A) -17 (B) 63
(C) 33 (D) 53

3. Simplify $5(2x - 3y) - 2[3x - (x - 4y)] + 7y$.

- (A) $6x - 2y$ (B) $2x - 16y$
(C) $6x - 16y$ (D) $6x + 16y$

4. Which expression is EQUIVALENT to $36 - 24x$?

(A) $-12(2x - 3)$

(B) $12(3 - 24x)$

(C) $-12(2x + 3)$

(D) $6(6 - 24x)$

5. A pattern of tiles has stage totals 4, 9, 16, 25, ... (each stage is a perfect square). The number of tiles ADDED when going from stage n to stage $(n + 1)$ is:

(A) $2n + 1$

(B) $n + 1$

(C) $2n + 3$

(D) $2n + 5$

6. In the expression $8 - 3x + 5x^2 - x^3$, list the coefficient of x , the coefficient of x^2 , and the constant term, in that order:

(A) 3, 5, 8

(B) -3, 5, -8

(C) -3, 5, 8

(D) -1, 5, 8

7. A rule says: a number x is first increased by 5, that result is doubled, and finally 3 times the original number is subtracted from it. The simplified expression is:

(A) $5x + 10$

(B) $-x + 5$

(C) $-5x + 10$

(D) $-x + 10$

8. Evaluate $(a - b)^2 - (a^2 - b^2)$ when $a = 7$ and $b = 3$.

(A) -8

(B) 24

(C) 40

(D) -24

9. Simplify $4a - \{2a - [3a - (a - 5)] - 4\}$.

(A) $4a - 1$

(B) $4a + 1$

(C) $4a + 9$

(D) $2a + 9$

10. Which expression is NOT equivalent to $30x - 45$?

(A) $15(2x - 3)$

(B) $5(6x - 8)$

(C) $-15(3 - 2x)$

(D) $5(6x - 9)$

11. A sequence has n th term $3n + 2$: the terms are 5, 8, 11, 14, The SUM of the 10th and the 20th terms is:

(A) 90

(B) 62

(C) 47

(D) 94

12. Four expressions are evaluated at $x = 4$: $P = x^2 - 3$, $Q = 3x + 1$, $R = 2x + 6$, $S = 4x - 5$. Exactly two of them are equal in value for this substitution. The common value of that equal pair is:

(A) 13

(B) 14

(C) 11

(D) 16

13. A hall is booked for a function: the base rent is 2000 rupees, each of g guests is charged 60 rupees, and a flat 500-rupee cleaning fee applies, but the host gets a 100-rupee discount for every full block of guests over the first 10 (assume the number over 10 is

exactly the variable part, charged at a net 60 - ... no discount applies within scope). Using g guests with the simple model 'base 2000 + 60 per guest + 500 cleaning', the total cost in rupees, simplified, is:

- (A) $60g + 2000$ (B) $60g + 2500$
(C) $2560g$ (D) $60g + 500$

14. Evaluate $(2x - y)^2$ divided by $(x + y)$ when $x = 3$ and $y = -1$.

- (A) 12.25 (B) 24.5
(C) 24 (D) 49

15. Simplify $(3x^2 - 2x + 5) - (x^2 - 4x - 3) + (2x - 1)$.

- (A) $2x^2 + 4x + 1$ (B) $2x^2 + 4x + 7$
(C) $2x^2 - 4x + 7$ (D) $4x^2 + 4x + 7$

16. Which expression is EQUIVALENT to $(\frac{1}{3})(9x - 6) - (\frac{1}{2})(4x - 10)$?

- (A) $x + 3$ (B) $x - 7$
(C) $5x + 3$ (D) $x - 3$

17. A staircase of cubes has layer counts 2, 6, 12, 20, 30, ... where the differences are 4, 6, 8, 10. A correct rule for the n th layer is $n(n + 1)$. The number of cubes ADDED going from layer 6 to layer 7 is:

- (A) 12 (B) 42
(C) 56 (D) 14

18. In the expression $-2x^2 + 7x - 9$, which single statement is entirely correct?

- (A) the coefficient of x^2 is 2, the coefficient of x is 7, and the constant term is 9
(B) the coefficient of x^2 is -2, the coefficient of x is -7, and the constant term is -9
(C) the coefficient of x^2 is -2, the coefficient of x is 7, and the constant term is -9
(D) the coefficient of x^2 is -2, the coefficient of x is 7, and the constant term is 9

19. A shopkeeper buys n crates of mangoes at 240 rupees each and twice as many crates of oranges at 180 rupees each, then pays a flat 150-rupee transport charge on the whole purchase. The total cost in rupees, simplified, is:

- (A) $420n + 150$ (B) $600n + 300$
(C) $600n + 150$ (D) $600n^2 + 150$

20. Evaluate $(a^2 - b^2)$ divided by $(a - b)$ when $a = 9$ and $b = 4$. (Compute numerator and denominator separately, then divide.)

- (A) 5 (B) 13
(C) 65 (D) 3.25

21. Simplify $3[2a - (a - b)] - 2[3b - (a + b)] + a$.

- (A) $6a + b$ (B) $4a - b$
(C) $6a - b$ (D) $6a - 5b$

22. Which expression is EQUIVALENT to $2(3x - 4) - 3(2x - 5) + 4(x - 2)$?

(A) $4x + 15$

(B) $4x - 21$

(C) $12x - 1$

(D) $4x - 1$

23. A theatre's seating grows: row 1 has 14 seats and each later row has 3 more seats than the row before. Using r for the row number, the total number of seats in the FIRST two rows combined, as an expression in r , is irrelevant; instead, the number of seats in row r is given by which rule, and what is it for row 8?

(A) $3r + 14$, giving 38

(B) $14r + 3$, giving 115

(C) $3r + 8$, giving 32

(D) $3r + 11$, giving 35

24. Four expressions are evaluated at $x = 5$: $A = x^2 - 10$, $B = 3x + 2$, $C = 2x + 1$, $D = 4x - 7$. Which ordering lists them from GREATEST to least value?

(A) A, B, C, D

(B) B, A, D, C

(C) B, A, C, D

(D) A, B, D, C

25. The number of diagonals of a polygon with s sides is given by the rule $s(s - 3)$ divided by 2. Using this rule, the number of diagonals of a polygon with 7 sides is:

(A) 28

(B) 7

(C) 10

(D) 14

26. Evaluate $-a^2 - (-a)^2 + 2a$ when $a = -3$. (Note: $-a^2$ means $-(a^2)$, while $(-a)^2$ squares the negative.)

(A) -24

(B) 0

(C) -6

(D) 12

27. Simplify $5xy - 3x^2 + 2yx - x^2 + 4 - xy$ and identify the coefficient of the xy -term in the result:

(A) 7

(B) 5

(C) 4

(D) 6

28. A mobile plan: a fixed monthly rental r rupees includes 100 free minutes, then 2 rupees per minute beyond 100. A user talks for m minutes in a month (m greater than 100). The monthly bill in rupees is:

(A) $2(m - 100) + r$

(B) $2m + r$

(C) $2m - 100 + r$

(D) $r(m - 100) + 2$

29. Evaluate $(3a - 2b)(a + b)$ by computing each bracket first and then multiplying, when $a = -1$ and $b = -2$.

(A) 3

(B) -3

(C) -9

(D) 9

30. A figure is built by joining n unit squares in a single row, then stacking one more identical row on top, forming a 2-by- n rectangle of unit squares. The perimeter of this rectangle, in units, simplified, is:

(A) $2n + 4$

(B) $2n + 2$

(C) $4n$

(D) $n + 4$

31. In the expression $5 - 2x + 3x^2 - 4x^3 + x^4$, the sum of ALL the coefficients (of x^0 through x^4 , each taken with its sign) equals which value, and this also equals the expression evaluated at which substitution?

(A) 15, equalling the value at $x = 1$

(B) 3, equalling the value at $x = -1$

(C) 3, equalling the value at $x = 1$

(D) -3, equalling the value at $x = 0$

32. A worker is paid w rupees for each ordinary hour and 1.5 times that rate for each overtime hour. In a week she works 40 ordinary hours and 6 overtime hours. Her total pay in rupees, simplified, is:

(A) $46w$

(B) $46.5w$

(C) $49w$

(D) $40w + 9$

33. Evaluate $(x + y)^2 - (x - y)^2$ when $x = 6$ and $y = 2.5$. (You may compute each square first.)

(A) 12.25

(B) 30

(C) 72.25

(D) 60

34. Simplify $2(a - 3b) - [3(a - b) - 2(2a - b)]$ and identify the coefficient of a in the result:

(A) 3

(B) 1

(C) -1

(D) 5

35. Which expression is EQUIVALENT to $(\frac{1}{2})(4x - 6) + (\frac{2}{3})(9x + 3) - 5x$?

(A) $3x + 1$

(B) $3x - 1$

(C) $8x - 1$

(D) $3x - 5$

36. A sequence has n th term $n^2 - n$: the terms are 0, 2, 6, 12, 20, The difference between the 8th term and the 6th term is:

(A) 14

(B) 26

(C) 12

(D) 30

37. A book costs b rupees and a pen costs p rupees. A gift pack contains 2 books and 3 pens; a deluxe pack contains 4 books and 1 pen. A store sells 5 gift packs and 2 deluxe packs. The total cost in rupees, simplified, is:

(A) $18b + 17p$

(B) $18b + 16p$

(C) $14b + 17p$

(D) $10b + 8p$

38. Evaluate $(2a)^2 - 2a^2 - a(a - 4)$ when $a = 3$. (Mind that $(2a)^2$ differs from $2a^2$.)

(A) 15

(B) 21

(C) 9

(D) 27

39. Simplify $7p - 2[3p - (p - q)] + 5q$ and state the result:

(A) $3p + 3q$

(B) $3p + 7q$

(C) $3p - q$

(D) $p + 3q$

40. Which expression is NOT equivalent to the perimeter of a rectangle of length $(2x + 1)$ and breadth $(x - 3)$?

(A) $2[(2x + 1) + (x - 3)]$

(B) $6x - 4$

(C) $2(3x - 2)$

(D) $6x + 4$

41. A pattern of matchsticks: figure 1 uses 6 sticks, figure 2 uses 11, figure 3 uses 16, increasing by 5 each time. The TOTAL number of sticks used in figures 1 through 4 together is:

(A) 44

(B) 21

(C) 54

(D) 60

42. In the expression $4x^3 - 0x^2 + 7x - 2$, a student claims 'the coefficient of x^2 is 0, so the x^2 term may simply be dropped from the written expression.' Which is the correct judgement?

(A) Wrong: dropping x^2 changes the degree of the expression

(B) Correct: a term with coefficient 0 contributes nothing, so $4x^3 + 7x - 2$ is the same expression

(C) Wrong: the constant -2 also becomes 0 once x^2 is removed

(D) Correct, but only if x is positive

43. A farmer plants x rows of wheat with 20 plants each and $(x + 4)$ rows of rice with 15 plants each. Then 30 plants in total are lost to pests. The number of surviving plants, simplified, is:

(A) $35x + 60$

(B) $35x + 30$

(C) $35x - 30$

(D) $300x + 30$

44. Evaluate $(a + 2b)^2 - 4b(a + b)$ when $a = 5$ and $b = 7$. (Hint: simplify the expression in letters first if you like.)

(A) 169

(B) 144

(C) 0

(D) 25

45. Simplify $(5a - 3b) - \{2a - [b - (a - b)]\}$ and state the result:

(A) $2a - b$

(B) $2a - 5b$

(C) $4a - b$

(D) $2a + b$

46. Which expression is EQUIVALENT to $3(2x - 1) - 2(x - 4) + (x - 3)$?

(A) $5x - 2$

(B) $5x + 2$

(C) $5x + 8$

(D) $7x + 2$

47. A sequence has n th term given by the rule $2n^2 - 1$: the terms are 1, 7, 17, 31, The 5th term MINUS the 3rd term equals:

- (A) 16
(C) 32

- (B) 48
(D) 24

48. Four expressions are evaluated at $x = 2$: $A = x^3$, $B = 4x$, $C = 2x + 5$, $D = x^2 + 1$. Exactly one pair has EQUAL value at this substitution. Which value is shared by that equal pair?

- (A) 8
(C) 5

- (B) 9
(D) 16

49. A delivery service: the first 5 packages in a batch cost 40 rupees each, and every package beyond the 5th costs 28 rupees each. For a batch of p packages (p greater than 5), the total cost in rupees, simplified, is:

- (A) $28p + 200$
(C) $40p - 60$

- (B) $28p + 60$
(D) $28(p - 5) + 40$

50. Evaluate the expression $(3x - 1)/2 - (x - 5)/4 + x$ when $x = 3$. (Substitute directly and mind the subtracted fraction.)

- (A) 6.5
(C) 4

- (B) 3.5
(D) 7.5

51. Simplify $4xy - 2x^2y + 3yx - x^2y - xy$ and state the fully combined expression:

- (A) $6xy - 2x^2y$
(C) $6xy - 3x^2y$

- (B) $5xy - 3x^2y$
(D) $3x^3y^2$

52. A magazine subscription costs a one-time 120-rupee joining fee plus a monthly charge. The monthly charge is 80 rupees for each of the first 6 months and 50 rupees for each month after the 6th. For a subscription lasting m months (m greater than 6), the total cost in rupees, simplified, is:

- (A) $50m + 480$
(C) $50m + 300$

- (B) $80m + 120$
(D) $50(m - 6) + 120$

53. Evaluate $a^2 + b^2$ given that $a + b = 5$ and $ab = 6$, using the relation $a^2 + b^2 = (a + b)^2 - 2ab$.

- (A) 25
(C) 11

- (B) 37
(D) 13

54. Simplify $6(2a - b) - 3[2a - (a - b)] - (a - 4b)$ and identify the coefficient of b in the result:

- (A) -2
(C) -9

- (B) 1
(D) -5

55. Which expression is EQUIVALENT to $2[3(x - 1) - (x - 4)] - (4x - 2)$?

- (A) $8x$
(C) 0

- (B) $4x + 4$
(D) 4

56. A pyramid of oranges has layer counts 1, 4, 9, 16, ... (square numbers from the top). The TOTAL number of oranges in the top 4 layers together is:

- (A) 30
- (B) 16
- (C) 25
- (D) 20

57. An expression is written as $3a + 5$ and a second as $3(a + 5)$. A student says 'these are equal because both contain 3 and 5 and the letter a.' For which value(s) of a are $3a + 5$ and $3(a + 5)$ actually equal?

- (A) For no value of a ($3a + 5 = 3a + 15$ would require $5 = 15$)
- (B) For every value of a (they are the same expression)
- (C) Only for $a = 0$
- (D) Only for $a = 5$

58. Evaluate $(2a - 3)(a + 2) - (a - 1)(a + 4)$ by computing each bracket first, when $a = -2$.

- (A) -6
- (B) 0
- (C) 6
- (D) 12

59. In the expression $9x^2 - 4$, a student writes it as $(3x - 2)(3x + 2)$ and checks by substitution. At $x = 2$, the value of $9x^2 - 4$ and the value of $(3x - 2)(3x + 2)$ are, respectively:

- (A) 32 and 32
- (B) 32 and 16
- (C) 36 and 32
- (D) 32 and 64

60. In the expression $6 + 5x - 2x^2$, a student writes the terms in order of increasing power as $-2x^2 + 5x + 6$ and claims this is a different expression because the terms moved. The correct judgement is:

- (A) Same expression: rearranging terms does not change the value for any x , so $-2x^2 + 5x + 6$ equals $6 + 5x - 2x^2$
- (B) Different expression: the leading term $-2x^2$ now makes it negative for all x
- (C) Different expression: the constant 6 must always be written first
- (D) Same value only when $x = 0$

Solutions — Mathematics (Class 7), Chapter 4: Expressions Using Letter-Numbers — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	C	31	C	41	C	51	C
2	B	12	A	22	D	32	C	42	B	52	C
3	C	13	B	23	D	33	D	43	B	53	D
4	A	14	B	24	B	34	A	44	D	54	D
5	C	15	B	25	D	35	B	45	A	55	D
6	C	16	A	26	A	36	B	46	B	56	A
7	D	17	D	27	D	37	A	47	C	57	A
8	D	18	C	28	A	38	B	48	A	58	C
9	C	19	C	29	B	39	A	49	B	59	A
10	B	20	B	30	A	40	D	50	D	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) First courier: $18(d - 3) + 70 = 18d - 54 + 70 = 18d + 16$. Second courier: $25d$. Extra cost of the second over the first = $25d - (18d + 16) = 25d - 18d - 16 = 7d - 16$. Writing $7d + 16$ fails to carry the minus onto the +16 constant. Adding the two per-km rates as $43d$ ignores that one whole cost is being subtracted from the other. Keeping -70 forgets that the 70 was already reduced by the 18×3 covered kilometres. Seven d minus sixteen is the simplified difference.

2. (B) $a^2 = 4$ so $2a^2 = 8$. $ab = (-2)(5) = -10$ so $-3ab = -3(-10) = +30$. $b^2 = 25$. Total $8 + 30 + 25 = 63$. Treating $-3ab$ as $-3(+10) = -30$ (dropping the negative sign on a) gives $8 - 30 + 25 = 3$, and a further slip lands on -17. Using $2a^2 = (2^2 - 2)^2 = 16$ instead of $2(a^2)$ gives $16 + 30 - \dots$ an inflated total. The value sixty-three is the only one that keeps every sign on the substitution.

3. (C) Inner bracket first: $3x - (x - 4y) = 3x - x + 4y = 2x + 4y$. Then $-2[2x + 4y] = -4x - 8y$. The first product $5(2x - 3y) = 10x - 15y$. Sum: $10x - 15y - 4x - 8y + 7y$. x-terms: $10x - 4x =$

6x. y-terms: $-15y - 8y + 7y = -16y$. Result $6x - 16y$. Distributing the -2 over only the $3x$ leaves the y-count short, giving $-2y$. Forgetting to combine the $10x$ and $-4x$ gives $2x$. Mis-signing the $-8y$ gives $+16y$. Six x minus sixteen y is correct.

4. (A) Factor out -12 : $-12(2x) + (-12)(-3) = -24x + 36 = 36 - 24x$. Check the choice: $-12(2x - 3) = -24x + 36$, correct. $12(3 - 24x) = 36 - 288x$, which over-shrinks the x-term. $-12(2x + 3) = -24x - 36$, wrong constant sign. $6(6 - 24x) = 36 - 144x$, again the wrong coefficient on x. Only minus-twelve times the quantity two x minus three reproduces the original.

5. (C) Stage n has $(n + 1)^2$ tiles, since stage $1 = 4 = 2^2$, stage $2 = 9 = 3^2$. Going to stage $n+1$ gives $(n + 2)^2$. The increase is $(n + 2)^2 - (n + 1)^2 = (n^2 + 4n + 4) - (n^2 + 2n + 1) = 2n + 3$. Check directly: $9 - 4 = 5 = 2(1)+3$, $16 - 9 = 7 = 2(2)+3$. The plain square-difference $2n + 1$ fits a pattern starting at 1^2 , not this one shifted up by one. $n + 1$ mistakes the side-length growth for the tile growth. Two n plus three is the actual difference.

6. (C) The term $-3x$ has coefficient -3 (the sign stays with the number). The term $5x^2$ has coefficient 5 . The constant term is the number with no letter, 8 . The order asked is coefficient of x , then x^2 , then constant: $-3, 5, 8$. Reporting $+3$ drops the minus that belongs to the x-term. Reporting -8 sign-flips a constant that is genuinely positive. Reporting -1 for x confuses the coefficient of x with that of x^3 . Minus three, five, eight is correct.

7. (D) Increase by 5 : $x + 5$. Double it: $2(x + 5) = 2x + 10$. Subtract 3 times the original: $2x + 10 - 3x = -x + 10$. Forgetting to distribute the doubling over the 5 (writing $2x + 5$) gives $-x + 5$. Doubling only x (and adding 5 separately as $5x$) misreads the wording. Minus x plus ten keeps both the distributed 10 and the single leftover x .

8. (D) $(a - b)^2 = (7 - 3)^2 = 4^2 = 16$. $a^2 - b^2 = 49 - 9 = 40$. So $16 - 40 = -24$. Treating $(a - b)^2$ as $a^2 - b^2$ (the classic squaring-a-difference error) would make the whole thing 0 , and a partial slip lands on -8 . Forgetting the leading minus on the bracket gives $+24$. Stopping at $a^2 - b^2$ reports 40 . Minus twenty-four keeps the true square 16 minus 40 .

9. (C) Innermost: $3a - (a - 5) = 3a - a + 5 = 2a + 5$. Next level: $[2a + 5]$ so $2a - (2a + 5) - 4 = 2a - 2a - 5 - 4 = -9$. Now $4a - \{-9\} = 4a + 9$. Mishandling the double minus as -9 kept negative gives $4a - 9$ then a slip to $4a - 1$. Dropping the -4 inside gives $4a + \dots$ a different constant. Four a plus nine results from turning the subtracted brace, which evaluates to negative nine, into a plus nine.

10. (B) Test each: $15(2x - 3) = 30x - 45$, equivalent. $-15(3 - 2x) = -45 + 30x = 30x - 45$, equivalent. $5(6x - 9) = 30x - 45$, equivalent. But $5(6x - 8) = 30x - 40$, which has the wrong constant (-40 not -45), so it is NOT equivalent. The trap rewards checking the constant term after distributing, not just the x-coefficient.

11. (D) 10th term: $3(10) + 2 = 32$. 20th term: $3(20) + 2 = 62$. Sum $32 + 62 = 94$. Using only $3n$ and forgetting the $+2$ on each term gives $30 + 60 = 90$. Reporting just the 20th term (62) ignores the word SUM. Treating the sum's index as 30 and computing one term $3(30) + 2 = 92$ is close but wrong logic; ninety-four is the genuine total of the two named terms.

12. (A) At $x = 4$: $P = 16 - 3 = 13$; $Q = 12 + 1 = 13$; $R = 8 + 6 = 14$; $S = 16 - 5 = 11$. The only equal pair is P and Q, both 13. $R = 14$ and $S = 11$ are the distractor values of the unmatched expressions. Computing x^2 as $(4)^2$ correctly is essential; a slip to 16 for P would wrongly pair it elsewhere. Thirteen is the shared value of P and Q.

13. (B) Add the two fixed charges: $2000 + 500 = 2500$. The only per-guest charge is 60 each, giving 60g. Total $60g + 2500$. Forgetting the 500 cleaning fee leaves $60g + 2000$. Merging the fixed 2500 into the rate as 2560g wrongly multiplies a one-time fee by g. Dropping the 2000 base leaves $60g + 500$. Sixty g plus twenty-five hundred combines both flat fees correctly.

14. (B) $2x - y = 6 - (-1) = 7$, so $(2x - y)^2 = 49$. $x + y = 3 + (-1) = 2$. So $49 / 2 = 24.5$. Computing $2x - y$ as $6 - 1 = 5$ (sign slip) gives $25/2 = 12.5$, near a wrong choice. Forgetting to square before dividing or halving twice yields 12.25. Twenty-four point five is the squared numerator over the correct denominator two.

15. (B) Distribute the middle minus over every term: $-(x^2 - 4x - 3) = -x^2 + 4x + 3$. Combine: x^2 terms $3 - 1 = 2$ giving $2x^2$; x terms $-2 + 4 + 2 = 4$ giving $4x$; constants $5 + 3 - 1 = 7$. Result $2x^2 + 4x + 7$. Distributing the minus over only the first term leaves a $+1$ constant. Keeping $-4x$ as $+(-4x)$... mishandling the $+4x$ flips the x -sign. Two x -squared plus four x plus seven is the full distribution.

16. (A) $(1/3)(9x - 6) = 3x - 2$. $(1/2)(4x - 10) = 2x - 5$. Subtract: $(3x - 2) - (2x - 5) = 3x - 2 - 2x + 5 = x + 3$. Failing to flip the sign of the -5 when subtracting the bracket gives $x - 7$. Adding the two leading coefficients as $5x$ ignores that one bracket is subtracted. Mis-signing the constants gives $x - 3$. x plus three is the careful difference.

17. (D) Layer 6: $6 \cdot 7 = 42$. Layer 7: $7 \cdot 8 = 56$. Added $= 56 - 42 = 14$. The differences run 4, 6, 8, 10, 12, 14 — the jump into layer 7 is the sixth difference, 14. Reading the difference pattern off by one gives 12. Reporting layer 6's total (42) or layer 7's total (56) answers a different question. Fourteen is the genuine increment.

18. (C) The sign always travels with the coefficient: $-2x^2$ gives coefficient -2 ; $+7x$ gives coefficient 7; the lone number -9 is the constant term. The option claiming $+2$ drops the minus on the squared term. The option with -7 sign-flips the x -term, which is genuinely positive. The option with $+9$ sign-flips the constant. Only minus two, seven, minus nine reads every sign faithfully.

19. (C) Mango cost: $240n$. Orange crates number $2n$ at 180 each: $180 \cdot 2n = 360n$. Subtotal $240n + 360n = 600n$, plus the one-time 150 transport = $600n + 150$. Forgetting to double the orange crates gives $240n + 180n = 420n$. Multiplying the flat charge by something doubles the constant to 300. Treating 'twice as many' as a power inflates to n -squared. Six hundred n plus one fifty is correct.

20. (B) Numerator $a^2 - b^2 = 81 - 16 = 65$. Denominator $a - b = 9 - 4 = 5$. $65 / 5 = 13$. (This equals $a + b = 13$, a useful check.) Reporting 5 gives only the denominator. Reporting 65 stops at the numerator. Dividing the wrong way ($5/65$... or mis-squaring) yields 3.25. Thirteen is the quotient and matches $a + b$.

21. (C) First brace: $2a - (a - b) = 2a - a + b = a + b$; times 3 gives $3a + 3b$. Second brace: $3b - (a + b) = 3b - a - b = -a + 2b$; times -2 gives $2a - 4b$. Add the trailing a . Total: $(3a + 2a + a) + (3b - 4b) = 6a - b$. Forgetting the minus on the second bracket flips to $+b$. Mis-distributing the leading 3 over only the first term loses $2a$. Six a minus b is the careful sum.

22. (D) $2(3x - 4) = 6x - 8$. $-3(2x - 5) = -6x + 15$. $4(x - 2) = 4x - 8$. x -terms: $6x - 6x + 4x = 4x$. Constants: $-8 + 15 - 8 = -1$. Result $4x - 1$. Distributing the minus over only the $2x$ (leaving $+5$ unchanged sign) gives constant $+15$. Forgetting one -8 lands on -21 . Adding all coefficients as $12x$ ignores the cancellation. Four x minus one is correct.

23. (D) Row 1 = 14 means at $r = 1$ the rule must give 14. With common difference 3 the rule is $3r + c$ where $3(1) + c = 14$, so $c = 11$: rule $3r + 11$. Check row 2: $3(2) + 11 = 17 = 14 + 3$, correct. Row 8: $3(8) + 11 = 35$. Writing $3r + 14$ over-counts because it gives 17 at row 1. Swapping rate and start gives $14r + 3$. Three r plus eleven, value thirty-five, fits the first row exactly.

24. (B) At $x = 5$: $A = 25 - 10 = 15$; $B = 15 + 2 = 17$; $C = 10 + 1 = 11$; $D = 20 - 7 = 13$. Ranking from greatest: B (17), A (15), D (13), C (11), so B, A, D, C . Assuming the squared expression is automatically largest puts A first wrongly. Mixing up C and D (11 vs 13) gives the orders ending C, D, B, A, D, C is the strict decreasing order.

25. (D) Substitute $s = 7$: $7(7 - 3)/2 = 7 \cdot 4/2 = 28/2 = 14$. Forgetting to halve gives 28. Computing $7 - 3$ then ignoring the multiply-and-halve gives a stray small number. Using $s(s - 3)$ without the division by 2 yields 28; only fourteen applies the full rule including the divide-by-two.

26. (A) $a = -3$. $a^2 = 9$, so $-a^2 = -9$. $(-a)^2 = (3)^2 = 9$, so $-(-a)^2 = -9$. $2a = -6$. Sum: $-9 - 9 - 6 = -24$. Treating $-a^2$ as $(+9)$ (mishandling the sign convention) cancels with the next term to give a wrong 0 or -6 . Recognising that BOTH $-a^2$ and $-(-a)^2$ equal -9 here, plus $2a = -6$, gives minus twenty-four.

27. (D) Note $2yx$ is the same like-term as xy . The xy -terms: $5xy + 2xy - xy = 6xy$, coefficient 6. (The x^2 terms combine to $-4x^2$ and the constant is $+4$, but the question asks only for the xy -coefficient.) Forgetting that $2yx = 2xy$ and treating it as unlike gives 4.

Adding the 2 twice gives 7. Missing the $-xy$ gives 7 or leaving 5. Six is the combined xy -coefficient.

28. (A) The rental r covers the first 100 minutes, so only $(m - 100)$ chargeable minutes cost 2 each: $2(m - 100) + r$. Writing $2m + r$ charges all m minutes, double-counting the 100 free ones. Writing $2m - 100$ distributes the 2 over only m and forgets to multiply the 100. Two times the quantity m minus one hundred, plus r , is correct.

29. (B) $3a - 2b = 3(-1) - 2(-2) = -3 + 4 = 1$. $a + b = -1 + (-2) = -3$. Product $1 * (-3) = -3$. Sign-slipping $-2b$ as $-(-2)$... computing $3a - 2b = -3 - 4 = -7$ (wrong sign on the $-2b$ term) leads astray. With both brackets correct, one times minus three is minus three.

30. (A) A 2-by- n rectangle has length n and width 2, so perimeter $= 2(\text{length} + \text{width}) = 2(n + 2) = 2n + 4$. Using a single row of n squares (perimeter $2n + 2$) ignores the second stacked row's effect on width. Treating it as 4 times n confuses area-like counting. Two n plus four is the perimeter of the two-high rectangle.

31. (C) Coefficients with signs: 5, -2, 3, -4, 1. Their sum $5 - 2 + 3 - 4 + 1 = 3$. Substituting $x = 1$ makes every power 1, so the expression equals exactly that coefficient sum, 3. Adding magnitudes without signs gives 15. The $x = 0$ value is just the constant 5, not the sum. Three, equalling the value at $x = 1$, is the correct pairing.

32. (C) Ordinary pay: $40w$. Overtime rate is $1.5w$ per hour for 6 hours: $1.5w * 6 = 9w$. Total $40w + 9w = 49w$. Forgetting the 1.5 multiplier and paying overtime at the plain rate gives $46w$. Multiplying 6 by 1.5 to get 9 but adding it as a constant 9 rather than $9w$ gives $40w + 9$. Forty-nine w combines both blocks at the right rates.

33. (D) $x + y = 8.5$, squared $= 72.25$. $x - y = 3.5$, squared $= 12.25$. Difference $72.25 - 12.25 = 60$. (This matches the identity $4xy = 4*6*2.5 = 60$.) Stopping at one square gives 72.25 or 12.25. Halving the result mistakenly gives 30. Sixty is the difference of the two squares and equals four $x y$.

34. (A) Inside the bracket: $3(a - b) = 3a - 3b$; $2(2a - b) = 4a - 2b$; so $3a - 3b - (4a - 2b) = 3a - 3b - 4a + 2b = -a - b$. The whole becomes $2(a - 3b) - [-a - b] = 2a - 6b + a + b = 3a - 5b$. Coefficient of a is 3. Forgetting to distribute the inner minus over $-2b$ gives a different a -coefficient. Mishandling the outer minus on the bracket gives 1. Three is the a -coefficient after full simplification.

35. (B) $(1/2)(4x - 6) = 2x - 3$. $(2/3)(9x + 3) = 6x + 2$. Sum so far: $2x - 3 + 6x + 2 = 8x - 1$. Subtract $5x$: $8x - 1 - 5x = 3x - 1$. Mis-adding the constants as $+1$ gives $3x + 1$. Forgetting to subtract the $5x$ leaves $8x - 1$. Dropping the $+2$ from the second product gives a -5 constant. Three x minus one is correct.

36. (B) 8th term: $8^2 - 8 = 64 - 8 = 56$. 6th term: $6^2 - 6 = 36 - 6 = 30$. Difference $56 - 30 = 26$. Using the consecutive difference (which would be $2n$ for going one step) instead of a

two-step gap gives 14. Reporting the 6th term itself gives 30. Twenty-six is the genuine gap between the 8th and 6th terms.

37. (A) Gift packs: 5 packs of $(2b + 3p) = 10b + 15p$. Deluxe packs: 2 packs of $(4b + 1p) = 8b + 2p$. Total: $(10b + 8b) + (15p + 2p) = 18b + 17p$. Forgetting the deluxe pens (2p) gives $18b + 15p$, or a near-miss 16p. Counting only one pack-type's books gives 14b. Eighteen b plus seventeen p adds both pack types fully.

38. (B) $(2a)^2 = (6)^2 = 36$. $2a^2 = 2 \cdot 9 = 18$. $a(a - 4) = 3(3 - 4) = 3(-1) = -3$. Expression: $36 - 18 - (-3) = 36 - 18 + 3 = 21$. Reading $(2a)^2$ as $2a^2 = 18$ (the classic squaring-the-coefficient slip) gives $18 - 18 + 3 = 3$ then drifts. Forgetting the double minus on $-a(a-4)$ gives $36 - 18 - 3 = 15$. Twenty-one keeps $(2a)^2 = 36$ and the +3 from the subtracted negative.

39. (A) Inner: $3p - (p - q) = 3p - p + q = 2p + q$. Times -2: $-4p - 2q$. Then $7p - 4p - 2q + 5q = 3p + 3q$. Distributing the -2 over only the 3p (leaving the +q alone) gives $3p + 7q$. Forgetting the inner sign flip on q gives $3p - q$. Three p plus three q is the careful result.

40. (D) Perimeter = $2(\text{length} + \text{breadth}) = 2[(2x + 1) + (x - 3)] = 2[3x - 2] = 6x - 4$. So $2[(2x+1)+(x-3)]$, $6x - 4$, and $2(3x - 2)$ all equal the perimeter. But $6x + 4$ has the wrong constant sign (the breadth's -3 should pull the constant negative), so it is NOT equivalent. The trap is dropping the minus on the breadth's constant when adding inside the bracket.

41. (C) Rule: $5n + 1$ (since $5(1)+1 = 6$, $5(2)+1 = 11$, $5(3)+1 = 16$). Figure 4: $5(4)+1 = 21$. Total of figures 1-4: $6 + 11 + 16 + 21 = 54$. Stopping at figure 3's running total or using figure 4 = $5(4) = 20$ (dropping the +1) gives 53 then drifts to 44. Reporting just figure 4 (21) answers a different question. Fifty-four is the sum of the first four figures.

42. (B) A term whose coefficient is 0 adds 0 for every value of x, so removing $0x^2$ leaves an identical expression: $4x^3 + 7x - 2$. The degree (highest power present with nonzero coefficient) is still 3, set by $4x^3$, so it is unchanged. The constant -2 is untouched by removing a different term. The truth of the simplification does not depend on the sign of x. The student is correct.

43. (B) Wheat: $20x$. Rice: $15(x + 4) = 15x + 60$. Subtotal $20x + 15x + 60 = 35x + 60$. Subtract the 30 lost: $35x + 60 - 30 = 35x + 30$. Forgetting to subtract the loss leaves $35x + 60$. Sign-confusing the 60 and 30 gives $35x - 30$. Thirty-five x plus thirty accounts for both crops and the loss.

44. (D) Expand in letters: $(a + 2b)^2 = a^2 + 4ab + 4b^2$; $4b(a + b) = 4ab + 4b^2$. Difference = a^2 . So the value is just $a^2 = 5^2 = 25$, regardless of b. Substituting blindly: $(5 + 14)^2 = 361$; $4 \cdot 7 \cdot (5+7) = 28 \cdot 12 = 336$; $361 - 336 = 25$, confirming. Stopping at $(a + 2b)^2$ alone gives 361, or other partial values give 169/144. Twenty-five, which is a^2 , is correct.

45. (A) Innermost: $b - (a - b) = b - a + b = -a + 2b$. Next: $2a - [-a + 2b] = 2a + a - 2b = 3a - 2b$. Outer: $(5a - 3b) - (3a - 2b) = 5a - 3b - 3a + 2b$. a-terms: $5a - 3a = 2a$. b-terms: $-3b + 2b = -b$. Result $2a - b$. Mis-signing the final bracket's $-2b$ (keeping it $-2b$) gives $-5b$. Forgetting to cancel an a gives $4a$. Two a minus b is correct.

46. (B) $3(2x - 1) = 6x - 3$. $-2(x - 4) = -2x + 8$. $(x - 3) = x - 3$. x-terms: $6x - 2x + x = 5x$. Constants: $-3 + 8 - 3 = 2$. Result $5x + 2$. Distributing -2 over only the x (leaving the -4 as -4) gives constant 2 from $-3 - 4 - 3$... a -10 slip, or $+8$ if the -3 's are mishandled. Adding all coefficients as $7x$ ignores cancellation. Five x plus two is correct.

47. (C) 5th term: $2(25) - 1 = 49$. 3rd term: $2(9) - 1 = 17$. Difference $49 - 17 = 32$. Using $2n - 1$ (linear, dropping the square) gives $9 - 5 = 4$ -type errors. Halving the gap gives 16 . Reporting the 5th term less something else gives 48 . Thirty-two is 49 minus 17 .

48. (A) At $x = 2$: $A = 2^3 = 8$; $B = 4 * 2 = 8$; $C = 4 + 5 = 9$; $D = 4 + 1 = 5$. The only equal pair is A and B , both 8 . $C = 9$ and $D = 5$ are the unmatched values. Reading x^3 as $3x = 6$ (the cube-versus-triple confusion) would break the match. Eight is the value shared by A and B .

49. (B) First 5 cost $5 * 40 = 200$. The remaining $(p - 5)$ cost 28 each: $28(p - 5) = 28p - 140$. Total $28p - 140 + 200 = 28p + 60$. Adding 200 without reducing it by the $28 * 5$ overlap gives $28p + 200$. Charging all p at 40 gives $40p$. Note $28(p - 5) + 40$ wrongly applies the flat-first-item pattern (only 40 for one item). Twenty-eight p plus sixty is the simplified two-rate total.

50. (D) At $x = 3$: $(3^3 - 1)/2 = 8/2 = 4$. $(x - 5)/4 = (3 - 5)/4 = -2/4 = -0.5$; subtracting it gives $-(-0.5) = +0.5$. Add $x = 3$. Total $4 + 0.5 + 3 = 7.5$. Forgetting that subtracting a negative adds 0.5 (instead treating it as -0.5) gives 6.5 . Dropping the $+x$ term gives 4.5 then 4 . Seven point five keeps the double-negative correct.

51. (C) xy -like terms: $4xy + 3yx - xy = 4 + 3 - 1 = 6$, giving $6xy$ (since $yx = xy$). x^2y -like terms: $-2x^2y - x^2y = -3x^2y$. The two groups are unlike (different powers of x) so stay separate: $6xy - 3x^2y$. Forgetting the lone $-x^2y$ leaves $-2x^2y$. Treating $3yx$ as unlike to xy gives $5xy$. Merging everything into one power is wrong since xy and x^2y are unlike. Six $x y$ minus three x -squared y is correct.

52. (C) Joining 120 . First 6 months: $6 * 80 = 480$. Remaining $(m - 6)$ months: $50(m - 6) = 50m - 300$. Total $120 + 480 + 50m - 300 = 50m + 300$. Adding $480 + 120 = 600$ without the -300 reduction gives $50m + 480$. Charging every month at 80 gives $80m + 120$. The option $50(m - 6) + 120$ forgets the 480 already-paid block. Fifty m plus three hundred is correct.

53. (D) $(a + b)^2 = 25$. $2ab = 12$. So $a^2 + b^2 = 25 - 12 = 13$. (Indeed $a, b = 2$ and 3 give $4 + 9 = 13$.) Reporting $(a+b)^2$ alone gives 25 . Adding $2ab$ instead of subtracting gives $25 + 12 = 37$. Thirteen uses the identity with the correct minus sign.

54. (D) Inner: $2a - (a - b) = a + b$; times -3 gives $-3a - 3b$. First product $6(2a - b) = 12a - 6b$. Last: $-(a - 4b) = -a + 4b$. Combine a-terms: $12a - 3a - a = 8a$. b-terms: $-6b - 3b + 4b = -5b$. So $8a - 5b$; coefficient of b is -5 . Forgetting the $+4b$ from the last bracket gives -9 . Mis-signing the inner bracket gives -2 . Minus five is the b -coefficient.

55. (D) Inside: $3(x - 1) = 3x - 3$; minus $(x - 4) = -x + 4$; sum $3x - 3 - x + 4 = 2x + 1$. Times 2 : $4x + 2$. Then $-(4x - 2) = -4x + 2$. Total: $4x + 2 - 4x + 2 = 4$ (the x -terms cancel). Forgetting to subtract the $4x$ leaves an $8x$. Mis-signing the final $-(-2)$ gives 0 instead of 4 . The constant four is what remains once the x -terms cancel.

56. (A) Layers: $1, 4, 9, 16$. Sum $= 1 + 4 + 9 + 16 = 30$. Reporting only the 4th layer (16) answers a different question. Stopping at the 3rd layer's running total ($1+4+9 = 14$) or mis-adding gives $20/25$. Thirty is the full sum of the first four square numbers.

57. (A) $3(a + 5) = 3a + 15$. Setting $3a + 5 = 3a + 15$ cancels $3a$ from both, leaving $5 = 15$, which is never true, so there is NO value of a making them equal. They are not the same expression — the 3 must distribute over the 5 . They differ by a constant 10 for every a , so $a = 0$ or $a = 5$ do not help either. For no value of a is the only correct statement.

58. (C) At $a = -2$: first pair $(2 \cdot -2 - 3) = -7$ and $(-2 + 2) = 0$, so the first product is $-7 \cdot 0 = 0$. Second pair $(a - 1) = -3$ and $(a + 4) = 2$, so the second product is $-3 \cdot 2 = -6$. Expression $= 0 - (-6) = +6$. Forgetting that subtracting -6 adds 6 gives -6 . Treating the second product as 0 too gives 0 . Six is the result of zero minus negative six.

59. (A) $9x^2 - 4$ at $x = 2$: $9 \cdot 4 - 4 = 36 - 4 = 32$. $(3x - 2)(3x + 2) = (6 - 2)(6 + 2) = 4 \cdot 8 = 32$. Both equal 32 , confirming the two expressions agree (a difference-of-squares pairing). Computing $9x^2$ as $(9x)^2$ inflates to wrong values. Mis-multiplying the brackets gives 16 or 64 . Thirty-two and thirty-two shows they are equal at this substitution.

60. (A) Addition is commutative, so the terms of an expression may be written in any order without changing its value for any substitution: $-2x^2 + 5x + 6$ is the very same expression as $6 + 5x - 2x^2$. The sign of the whole expression is not fixed by the leading term's sign (e.g. at $x = 1$ it equals 9 , positive). There is no rule forcing the constant first. The equality holds for every x , not only $x = 0$. They are the same expression.